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Probability Theory Paper¹

PROBABILITY THEORY AND TWO-WAY VENN-DIAGRAMS

QUESTION 1

1.1 Given: $P(M) = 0,45$

$$P(N) = 0,3$$

$$P(M \text{ or } N) = 0,615$$

1.1.1 Determine

(a) $P(M \text{ and } N)$. (2)

(b) $P(M \text{ and (not } N))$. (2)

(c) $P((\text{not } M) \text{ or (not } N))$. (2)

1.1.2 Are the events M and N mutually exclusive? Motivate. (2)

1.1.3 Are the events M and N independent? Motivate. (2)

QUESTION 2

2.1. Events A and B are independent of one another, with $P(A) = 0,4$ and $P(A \text{ and } B) = 0,12$.

Determine:

2.1.1 $P(B)$ (2)

2.1.2 $P(A \text{ or } B)$ (2)

2.1.3 $P(\text{not } A \text{ and not } B)$ (3)

2.2 $P(A) = \frac{1}{4}$ and $P(A \text{ or } B) = \frac{1}{3}$. Determine $P(B)$, as a simplified fraction, if:

2.2.1 A and B are mutually exclusive events. (3)

2.2.2 A and B are independent events. (5)

2.3 If $P(\text{not } A) = 0,7$ and $P(B) = 0,5$ determine $P(A \text{ or } B)$ if

2.3.1 A and B are mutually exclusive events (2)

2.3.2 A and B are independent events (3)

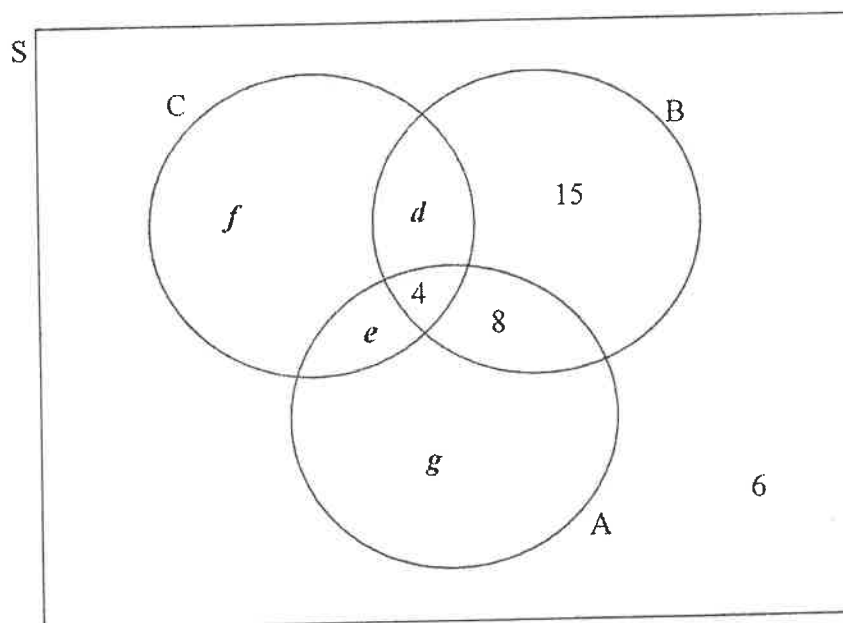
THREE - WAY VENN - DIAGRAMS

QUESTION 3

3.1 Research was conducted about driving under the influence of alcohol. Information obtained from traffic authorities in 54 countries on the methods that are used to measure alcohol levels in a person, are summarised below:

- 4 countries use all three methods (A, B and C).
- 12 countries use the alcohol content of breath (A) and blood-alcohol concentration (B).
- 9 countries use blood-alcohol concentration (B) and certificates issued by doctors (C).
- 8 countries use the alcohol content of breath (A) and certificates issued by doctors (C).
- 21 countries use the alcohol content of breath (A).
- 32 countries use blood-alcohol concentration (B).
- 20 countries use certificates issued by doctors (C).
- 6 countries use none of these methods.

Below is a partially completed Venn diagram representing the above information.



3.1.1 Use the given information and the Venn diagram to determine the values of d , e , f and g . (4)

3.1.2 For a randomly selected country, calculate:

- (a) $P(A \text{ and } B \text{ and } C)$ (1)
- (b) $P(A \text{ or } B \text{ or } C)$ (1)
- (c) $P(\text{only } C)$ (1)
- (d) $P(\text{that a country uses exactly two methods})$ (1)

QUESTION 3.2-

At Smartlearn High School there are 320 learners in Grade 12, 270 of whom take one or more of Mathematics, History and Economics. The following is known about the number of learners who take these subjects:

- 70 take History
- 50 take Economics
- 200 take Mathematics
- 20 take Mathematics and History
- 10 take History and Economics
- 25 take Mathematics and Economics
- x learners take Mathematics and History and Economics

- 3.2.1 Represent the information above in a Venn-diagram.
Represent Mathematics with M, History with H and Economics with E. (8)
- 3.2.2 Determine the number of learners, x , who take Mathematics, History and Economics. (2)
- 3.2.3 Determine $P(\text{not } (M \text{ or } H \text{ or } E))$ and state in words what your answer means. (2)
- 3.2.4 Determine the probability that a learner takes at least two of these subjects. (3)

TREE DIAGRAMS

QUESTION 4

City crime records show that 20% of all crimes are violent. 90% of violent crimes are reported versus 70% of non-violent crimes.

- 4.1 Represent the above by means of a tree diagram, showing all possible outcomes and their associated probabilities. (4)
- 4.2 What is the overall reporting rate for crimes in the city? I.e. what percentage of crimes are reported? (3)

QUESTION 5

A box contains 7 red cubes and 8 yellow cubes. Two cubes are taken out one at a time without looking at the colour. The cubes are not replaced.

- 5.1 Draw a tree diagram showing all possible outcomes. (4)
- 5.2 Find the probability that
- 5.2.1 both cubes removed were yellow. (3)
- 5.2.2 the two cubes were of different colours. (3)

TWO WAY CONTINGENCY TABLES

QUESTION 6

The following table shows 100 people's gender vs. their consumption of ice-cream:

	Consumes lots of ice-cream	Consumes little ice-cream	TOTAL
Male	15	?	45
Female	?	20	55
TOTAL	?	?	100

- 6.1 If a person is chosen randomly from these 100, what is the probability that the person
- 6.1.1 is female?
 - 6.1.2 consumes lots of ice-cream?
 - 6.1.3 is female and consumes lots of ice-cream?
- 6.2 Are the events {female} and {consumes lots of ice cream} independent? Motivate.

QUESTION 7

A survey concerning their holiday preferences was done with 180 staff members. The options they could choose from were to:

- Go to the coast
- Visit a game park
- Stay at home

The results were recorded in the table below:

	Coast	Game Park	Home	Total
Male	46	24	13	83
Female	52	38	7	97
Total	98	62	20	180

- 7.1 Determine the probability that a randomly selected staff member:
- 7.1.1 Is male (1)
 - 7.1.2 Does not prefer visiting a game park (2)
- 7.2 Are the events 'being a male' and 'staying at home' independent events. Motivate your answer with relevant calculations. (4)

COUNTING PRINCIPLE

QUESTION 8

Nametso may choose DVDs from three categories as listed in the table below:

Drama	Romance	Comedy
• <i>Last Hero</i> • <i>Midnight</i> • <i>Stranger Calls</i> • <i>Missing in Action</i> • <i>Only 40 Seconds Left</i>	• <i>One Heart</i> • <i>You and Me</i> • <i>Love Song</i> • <i>Bird's First Nest</i>	• <i>Laughing Dragon</i> • <i>Falling Down</i> • <i>Sitting on the Stairs</i>

- 8.1 Nametso must choose ONE DVD from the Drama category. What is the probability that she will choose *Midnight*? (2)
- 8.2 How many different selections are possible if her selection must include ONE drama, ONE romance and ONE comedy? (2)
- 8.3 Calculate the probability that she will have *Last Hero* and *Laughing Dragon* as part of her selection in QUESTION 10.2.2. (2)

QUESTION 9

Another school, Tulani High, has a sports awards ceremony. Tulani High has a basketball team consisting of 5 players and a volleyball team consisting of 6 players.

- 9.1 All the basketball players sit in a single row at the ceremony. There are no restrictions on who sits in which position. In how many different ways can they be seated? (1)
- 9.2 The decision is taken that the captain must sit in the first seat of the row. The two vice-captains have to be seated next to each other in any of the remaining seats. In how many different ways can the basketball players be seated now? (3)
- 9.3 After the interval, the basketball team and the volleyball team sit in the same row at the ceremony. Calculate the probability that the basketball players will sit together and the volleyball players will sit together. Assume that seating positions are allocated randomly. Give your answer as a simplified fraction. (3)

QUESTION 10

5 girls and 3 boys are to sit on a bench in a straight line.

- 10.1 In how many ways can they all sit on the bench? (3)
- 10.2 In how many ways can they all sit on the bench with girls seated together? (2)
- 10.3 Determine the probability that all girls are seated next to each other. (2)

QUESTION 11

- 11.1 A password consists of five different letters of the English alphabet. Each letter may be used only once. How many passwords can be formed if:
 - 11.1.1 All the letters of the alphabet can be used (2)
 - 11.1.2 The password must start with a 'D' and end with an 'L' (2)
- 11.2 Seven cars of different manufacturers, of which 3 are silver, are to be parked in a straight line.
 - 11.2.1 In how many different ways can ALL the cars be parked? (2)
 - 11.2.2 If the three silver cars must be parked next to each other, determine in how many different ways the cars can be parked. (3)

QUESTION 12

Consider the word: EINSTEIN.

- 12.1 How many different letter arrangements are possible if all the letters are used? (2)
- 12.2 What is the probability that identical letters will be grouped together? (4)

Probability Theory: Solutions

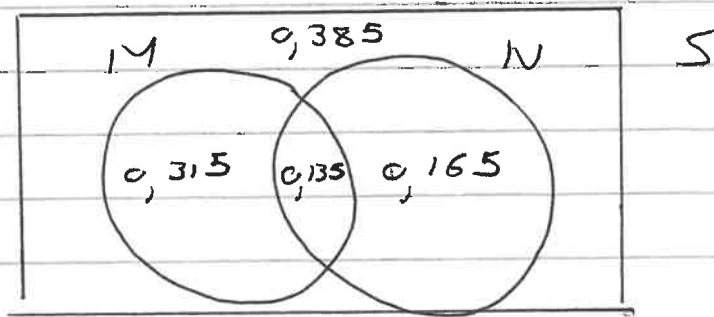
Question 1:

1.1.1. a) $P(M \text{ or } N) = P(M) + P(N) - P(M \text{ and } N)$

$$0,615 = 0,45 + 0,3 - P(M \text{ and } N)$$

$$\underline{P(M \text{ and } N) = 0,135}$$

b)



$$P(M \text{ and (Not } N))$$

$$= \underline{0,315}$$

c) $P(\text{Not } M \text{ or Not } N)$

$$= 0,315 + 0,385 + 0,165$$

$$= \underline{0,865}$$

1.1.2. No; $P(N \text{ and } M) \neq 0$

1.1.3 $P(M) \cdot P(N) = 0,45 \times 0,3$

$$= 0,135$$

$$= P(M \text{ and } N)$$

$\therefore M$ and N are independent

Question 2:

$$2.1.1. \quad P(A), P(B) = P(A \text{ and } B)$$

$$0,4 \cdot P(B) = 0,12$$

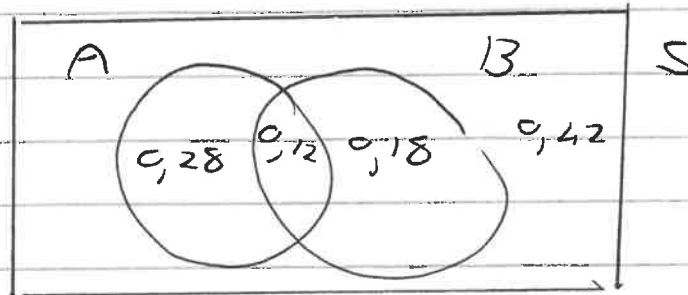
$$\underline{P(B) = 0,3,}$$

$$2.1.2. \quad P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$= 0,4 + 0,3 - 0,12$$

$$= \underline{0,58,}$$

2.1.3



$$P(\text{not } A \text{ and } \text{not } B) = \underline{0,42,}$$

$$2.2.1. \quad P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\frac{1}{3} = \frac{1}{4} + P(B) - 0$$

$$\underline{\frac{1}{12} = P(B),}$$

$$2.2.2. \quad P(A \text{ or } B) = P(A) + P(B) - P(A) \cdot P(B)$$

$$\frac{1}{3} = \frac{1}{4} + P(B) - \frac{1}{4} P(B)$$

$$\frac{1}{12} = \frac{3}{4} P(B)$$

$$\underline{P(B) = \frac{1}{9},}$$

$$2.3.1. \quad P(A) = 1 - P(\text{not } A)$$

$$= 1 - 0,7$$

$$= \underline{0,3}$$

$$\begin{aligned}
 P(A \text{ or } B) &= P(A) + P(B) \\
 &= 0,3 + 0,5 \\
 &= \underline{0,8}
 \end{aligned}$$

$$\begin{aligned}
 2.3.2 \quad P(A \text{ or } B) &= P(A) + P(B) - P(A) \cdot P(B) \\
 &= 0,3 + 0,5 - (0,3)(0,5) \\
 &= \underline{0,65}
 \end{aligned}$$

Question 3

$$\begin{aligned}
 3.1.1. \quad d &= 9 - 4, & e &= 8 - 4 \\
 &= \underline{5}, & &= \underline{4},
 \end{aligned}$$

$$\begin{aligned}
 f &= 20 - 5 - 4 - 4, & g &= 21 - 4 - 8 - 4 \\
 &= \underline{7}, & &= \underline{5},
 \end{aligned}$$

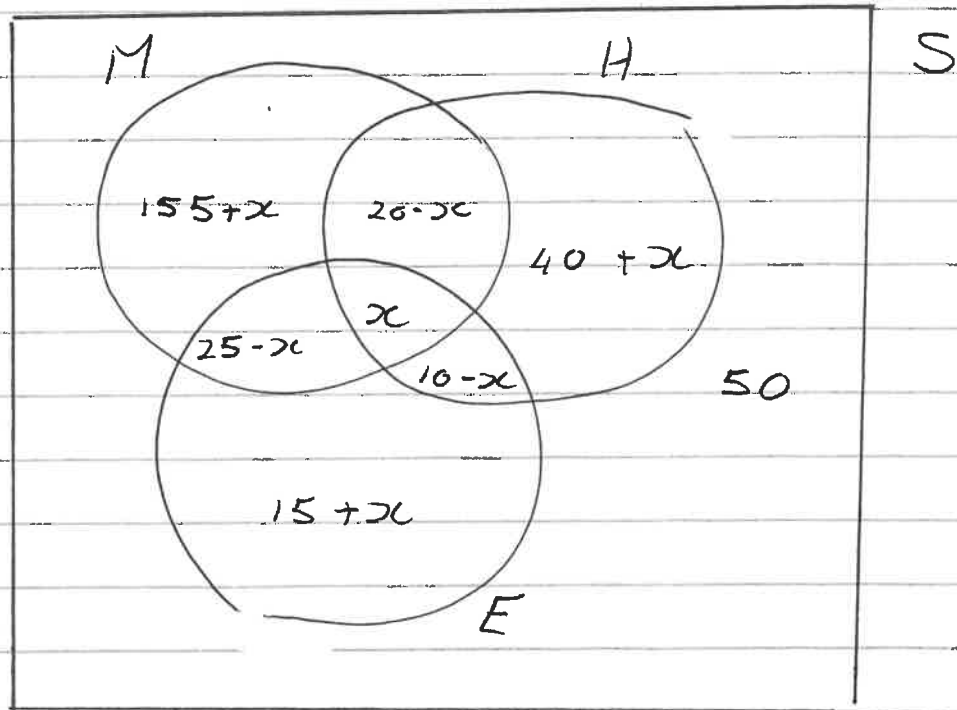
$$\begin{aligned}
 3.1.2. \quad a) \quad P(A \text{ and } B \text{ and } C) \\
 &= \frac{4}{54} \\
 &= \frac{2}{27},
 \end{aligned}$$

$$\begin{aligned}
 b) \quad P(A \text{ or } B \text{ or } C) \\
 &= \frac{48}{54} \\
 &= \frac{8}{9},
 \end{aligned}$$

$$\begin{aligned}
 c) \quad P(\text{only } C) \\
 &= \frac{7}{54},
 \end{aligned}$$

$$\begin{aligned}
 & P(\text{exactly two methods}) \\
 &= \frac{4 + 5 + 8}{54} \\
 &= \frac{17}{54}
 \end{aligned}$$

3.2.1.



$$3.2.2. \quad 155 + x + 25 - x + x + 20 - x + 40 + x + 10 - x + 15 + x + 50 = 320$$

$$315 + x = 320$$

$$x = 5$$

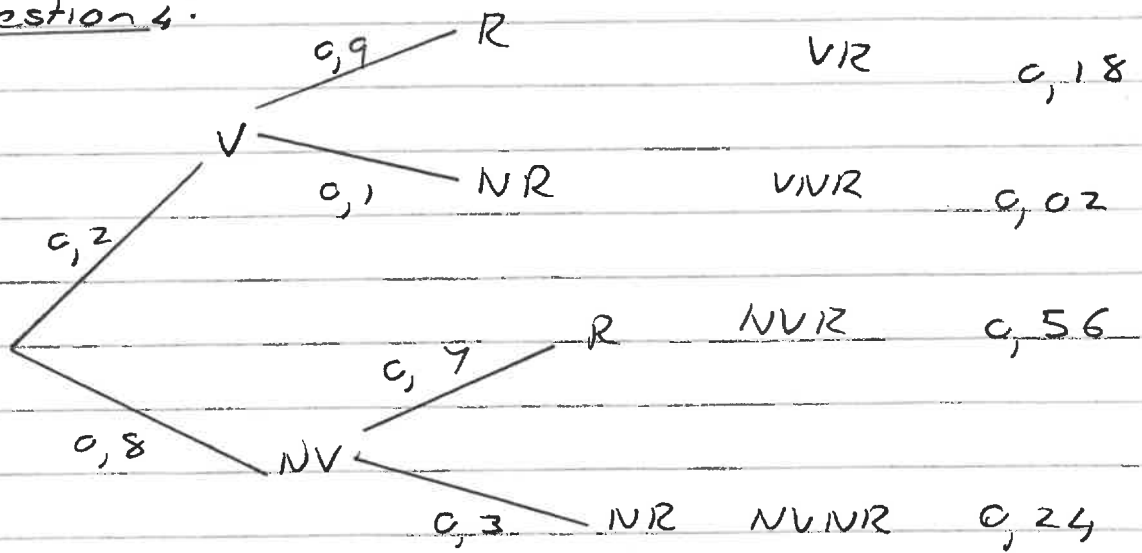
$$3.2.3. \quad \frac{50}{320} = \frac{5}{32}$$

The probability of a learner taking none of these subjects

$$3.2.4. \quad \frac{15 + 5 + 20 + 5}{320} = \frac{9}{64}$$

Question 4:

4.1.



4.2 $P(\text{Reported})$

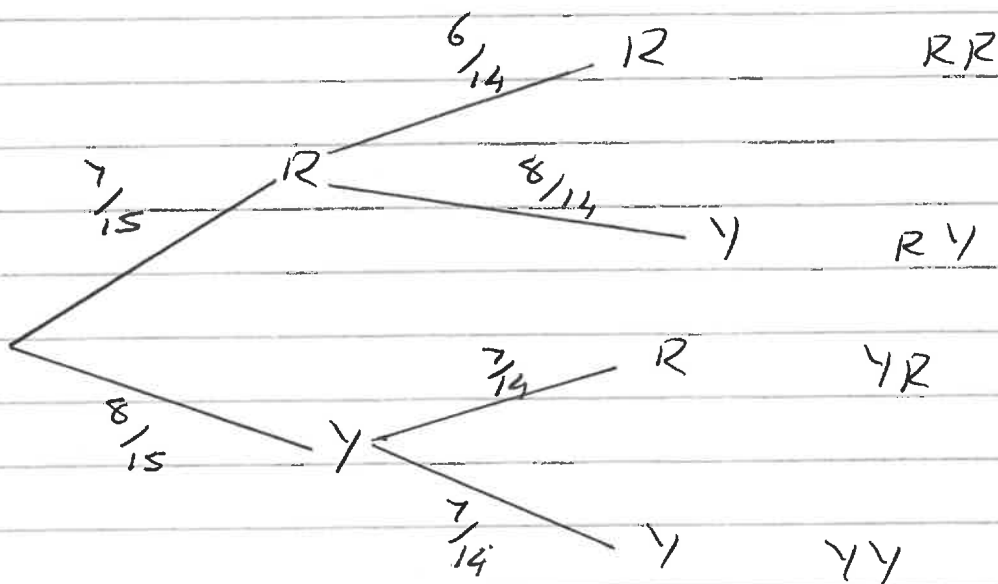
$$= P(NR) + P(NVR)$$

$$= 0,18 + 0,56$$

$$= \underline{0,74}$$

Question 5:

5.1.



5.2.1.

$$P(YY) = \frac{8}{15} \times \frac{7}{14}$$
$$= \underline{\frac{4}{15}}$$

5.2.2.

$$P(YR) + P(RY)$$

$$= \frac{8}{15} \times \frac{7}{14} + \frac{7}{15} \times \frac{8}{14} = \underline{\frac{8}{15}}$$

(6)

Question 6:

$$6.1.1. \quad P(\text{Female}) = \frac{55}{100} = \frac{11}{20},$$

$$6.1.2. \quad P(\text{Lots of Ice-cream}) = \frac{45}{100} = \frac{9}{20},$$

$$6.1.3. \quad P(\text{Female and Lots of Ice-cream})$$

$$= \frac{35}{100}$$

$$= \frac{7}{20}$$

$$6.2. \quad P(F) \cdot P(LI) \qquad P(F \text{ and } LI)$$

$$= \frac{11}{20} \times \frac{9}{20} \qquad = 0,35$$

$$= \frac{99}{400}$$

$$= 0,2475$$

$$\therefore P(F \text{ and } LI) \neq P(F) \cdot P(LI)$$

$\therefore$ Events are not independent

Question 7:

$$7.1.1. \quad P(\text{Male}) = \frac{83}{180}$$

$$7.1.2. \quad P(\text{not game park}) = \frac{98 + 20}{180} = \frac{59}{90}$$

$$7.2. \quad P(M), \quad P(H)$$

$$= \frac{83}{180} > \frac{20}{180}$$

$$= \frac{83}{1620}$$

$$= \underline{0.0512}$$

$$P(M \text{ and } H)$$

$$= \frac{13}{180}$$

$$= \underline{0.0722}$$

$$\therefore P(M \text{ and } H) \neq P(M) \cdot P(H)$$

$\therefore$ events are not independent.

Question 8

8.1.

$$\frac{1}{5}$$

8.2

$$5 \times 4 \times 3$$

$$= \underline{60}$$

8.3

$$\frac{1 \times 4 \times 1}{60}$$

$$= \frac{1}{15}$$

Question 9

9.1.

$$5! = \underline{120 \text{ ways}}$$

9.2

$$1 \times 3! \times 2! = \underline{12 \text{ ways}}$$

9.3

$$\frac{2! \times 5! \times 6!}{11!} = \frac{1}{231}$$

Question 10:

$$10.1. \quad 8! = \underline{40320},$$

$$10.2. \quad 5! \times 4! = \underline{2880},$$

$$10.3. \quad \underline{2880}$$

$$40320$$

$$= \underline{1}$$

$$\underline{14}$$

Question 11:

$$11.1.1. \quad 26 \times 25 \times 24 \times 23 \times 22$$

$$= \underline{7893600},$$

$$11.1.2. \quad 1 \times 24 \times 23 \times 22 \times 1$$

$$= \underline{12144},$$

$$11.2.1. \quad 7! = \underline{5040},$$

$$11.2.2. \quad 3! \cdot 5! = \underline{720},$$

Question 12:

$$12.1. \quad \begin{array}{ccccc} E & I & V & S & T \\ E & I & V & & \end{array}$$

$$E \quad I \quad V$$

$$\underline{8!} = 5040$$

$$2! \cdot 2! \cdot 2!$$

$$12.2. \quad \frac{5!}{5040} = \frac{1}{42}$$